

Stereotactic radiosurgery for brain tumours

Stereotactic radiosurgery (SRS) is a specialised way of giving radiotherapy in 1 high dose. It may be used to treat secondary brain tumours (brain metastasis) or small primary brain tumours.

On this page

- [What is stereotactic radiosurgery \(SRS\)?](#)
- [Radiotherapy machines for SRS](#)
- [Planning SRS treatment for brain tumours](#)
- [Having SRS treatment for brain tumours](#)
- [Side effects of SRS for brain tumours](#)
- [How a brain tumour may affect your right to drive](#)
- [After SRS for brain tumours](#)
- [How we can help](#)

What is stereotactic radiosurgery (SRS)?

Stereotactic radiosurgery (SRS) is a way of giving [radiotherapy](#) to a [brain tumour](#). It is given as a single high dose of radiotherapy, to a small area. Sometimes it may be divided into smaller doses given over a number of days. This is called stereotactic radiotherapy (SRT). Each daily dose is called a fraction. Stereotactic radiosurgery does not involve any surgery.

There are different ways of giving SRS. SRS can be given as 1 or more beams that move in an arc shape around the tumour. Other ways include using many small, thin radiotherapy beams given from different angles. These beams cross over at the tumour.

Giving SRS in this way means the dose is focused within the tumour, while limiting the dose to the surrounding areas.

SRS is not suitable for everyone with a brain tumour. It is usually only used to treat small tumours that are not close to important structures in the brain. Whether you can have SRS may depend on:

- the size of the tumour
- where the tumour is in [the brain](#)
- the [type of tumour](#)
- the number of tumours in the brain.

SRS can be used to treat brain metastases. These are brain tumours which have spread from cancers in other parts of the body. SRS can be given as a main treatment, or in combination with surgery.

SRS can also be used to treat some small non-cancerous (benign) primary brain tumours such as:

- vestibular schwannoma (also called [acoustic neuroma](#))
- [meningioma](#)
- [pituitary_gland tumour](#).

Radiotherapy machines for SRS

There are different types of radiotherapy machine which can deliver SRS. These include:

- Gamma Knife™
- CyberKnife™
- Linear accelerator (LINAC).

This treatment is not available at every hospital. You may have to travel to have this treatment. You may be able to get [help with travel costs](#).

Planning SRS treatment for brain tumours

Before you have SRS, it needs to be planned. This is an important part of your treatment. Planning makes sure the radiotherapy is aimed precisely at the tumour and avoids nearby healthy areas of the brain as much as possible.

Your treatment may be planned on the same day as it is given, or at an earlier appointment. Your radiotherapy team will tell you what to expect.

Masks and headframes

When having radiotherapy to your brain, you will be asked to wear a radiotherapy mask or headframe. This helps to hold your head and neck still and in exactly the right position. This will help make your treatment as accurate and effective as possible.

Masks

Masks are usually made from plastic mesh that is moulded to fit the shape of your face. You may have a mask made on the same day that your treatment is planned, or at an earlier appointment.

For your treatment, you lie on a treatment couch. The mask is gently placed over your face and fixed to the couch. The mask should be tight but not uncomfortable. You can breathe normally while you are wearing it.

We have more information about [how radiotherapy masks are made](#).

Head frames

For some types of SRS, you wear a lightweight plastic or metal head frame. If you are having SRT, you will not need a head frame.

The frame sits around your head and stays firmly in place by pressing against your skull at 4 points. This sounds scary and it may be a bit uncomfortable, but it should not be painful. You have local anaesthetic to numb the 4 points where the frame will push into your skin. Your team gently position the frame around your head.

Head frames are usually fitted on the same day that you have SRS.

Some types can be fitted, then taken off until your treatment session. Others are fixed. This means you wear the frame all day until your treatment session is over. You can move around with it on and can eat and drink normally. It can help to wear tops that button or zip up, because it can be difficult to take clothes off over your head.

For your treatment, the frame is fixed to the treatment couch. At the end of your treatment session your team gently removes the frame from your head. The frame will leave 4 tiny wounds on your skin. These usually heal in a few days. Your radiotherapy team will give you information about how to care for these small wounds. If the wounds feel sore, ask them about painkillers.

Support

You may feel very nervous about wearing a mask or head frame. You may want to bring some calming music or a relaxation podcast to listen to during your treatment. Most people cope well with the support of the radiotherapy team. If you are worried or uncomfortable, tell them, so they can help you.

Scans and planning

Before you have your treatment, you have a [brain CT scan](#) (as part of radiotherapy planning) and a [brain MRI scan](#). Your radiotherapy team uses information from the scans to plan your treatment.

The radiotherapy team helps you get into position for the scan. You lie on a treatment couch in the same position you will be in for treatment. The mask or head frame is gently fixed to the couch, so your head and neck do not move during the scan. If you are uncomfortable, tell the staff. They may be able to adjust the mask or frame slightly or help you move into a more comfortable position.

During the scan you may be alone in the room. Your radiotherapy team can see, hear and talk to you, and are close by if you need them at any time.

Before your treatment

Your radiotherapy team will talk to you about any medications you take. It is important to tell them if you have any allergies. If you take painkillers, they may suggest you take them before your treatment so that you are comfortable.

Radiotherapy can cause some swelling of the area being treated. This can cause side effects or make any symptoms of the brain tumour worse for a time. Your doctor will give you [steroids](#) to take before your treatment to help this. They may also give you medicines to help reduce the risk of a seizure (fit). You usually keep taking them for a short time after finishing treatment. Follow your doctor's instructions carefully and do not stop taking the steroids without your doctor's advice.

Having SRS treatment for brain tumours

For your SRS treatment, you lie in the same position as you did for the planning scans. The radiotherapy team will help you get comfortable and will fix the mask or head frame to the treatment couch.

The treatment itself is not painful. You may hear a slight buzzing noise from the radiotherapy machine during treatment. Depending on the type of radiotherapy machine, the machine may move around you during the treatment. The couch you lie on may change position slightly during treatment.

In some treatment rooms you can listen to music to help you relax. If you would like to listen to your own music, ask your radiotherapy team if this is possible.

Stereotactic radiosurgery is usually given in 1 session of treatment. Occasionally it may be broken up with short breaks or spread over several days (SRT). A session of treatment can take from about 15 minutes up to 4 hours. Sometimes, it may take longer. This depends on the type of machine used and the number of tumours being treated. Your radiotherapy team will tell you what to expect. Tell them if you are worried about any part of having treatment. They will try to make things as comfortable as possible for you.

The team will not be in the same room with you when you have the treatment. But they can see, hear and talk to you, and will be close by if you need them at any time. You can usually go home when the treatment is finished.

SRS does not make you radioactive. It is safe for you to be with other people, including children, after your treatment.

In this video, Esther, a Lead Stereotactic Radiographer, explains what stereotactic radiosurgery is, how it works and how it can be used to treat brain tumours.

Side effects of SRS for brain tumours

Side effects can depend on which area of the brain has been treated and the dose of radiation given. Your doctor will explain the likely side effects of your treatment. Some side effects are mild and easy to cope with. Others may be managed with drugs or other treatments.

You may develop some of the side effects mentioned. You are unlikely to have them all. Side effects may include tiredness, headaches and feeling sick. We have more information about [side effects after radiotherapy for a brain tumour](#).

Side effects usually get better gradually after your treatment. Sometimes side effects or symptoms of the brain tumour get worse for a while after treatment has finished. If this happens, you may feel upset or worry that treatment is not working. But this is usually a normal reaction to the radiotherapy. It may also happen because your steroids have been reduced or stopped.

Sometimes healthy cells in the treated area can become damaged from the treatment. This is called radiation necrosis. This can happen years after having the treatment. These may include headaches, feeling drowsy or memory loss. Symptoms of this depend on the area that was treated. Your specialist doctor or nurse can tell you more about this.

Tell your radiotherapy team about any side effects. They may give you drugs to help and practical advice about managing side effects.

How a brain tumour may affect your right to drive

Following diagnosis and treatment for a brain tumour, most people will not be allowed to drive for a period of time. If you drive it is important to discuss with your doctor how your diagnosis and treatment for a brain tumour affects your right to drive.

If you have a driving licence, you must tell the licencing agency (DVLA or DVA) that you have been diagnosed with a brain tumour. We have more information about how a [brain tumour may affect your right to drive](#).

After SRS for brain tumours

After treatment, you will have regular check-ups with your doctor and nurse. Sometimes SRS causes side effects that develop months or even years later. These are called late effects.

Your radiotherapy team will explain your risk of these before you agree ([consent](#)) to have SRS. You can use your follow-up appointments to talk to your doctor about any side effects, or other worries you have. They can tell you about ways to help manage the side effects.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our primary brain tumour information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

EANO-ESMO Clinical Practice Guidelines for prophylaxis, diagnosis, treatment and follow-up: Neurological and vascular complications of primary and secondary brain tumours. 2021. Available from www.eano.eu/publications/eano-guidelines/eano-esmo-clinical-practice-guidelines-for-prophylaxis-diagnosis-treatment-and-follow-up-neurological-and-vascular-complications-of-primary-and-secondary-brain-tumours [accessed August 2024].

NICE Guideline NG99. Brain tumours (primary) and brain metastases in over 16s. 2018 (updated 2021). Available from: www.nice.org.uk/guidance/ng99 [accessed August 2024].

[Cancer Information Development team](#)

Author

[Professor Catherine McBain](#)

Reviewer

Consultant Clinical Oncologist

Date reviewed

Reviewed: 01 May 2024 | Next review: 01 May 2027

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

